

GCSE Mathematics

Predicted Paper 2 — Higher Tier

Set E

Pearson Edexcel style (1MA1 / 2H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. A meal costs £45 before service charge. A service charge of 12.5% is added. Work out the total cost of the meal including the service charge.

(2 marks)

2. Alice, Ben and Chen share £540 in the ratio 2 : 3 : 4. Work out how much each person receives.

(3 marks)

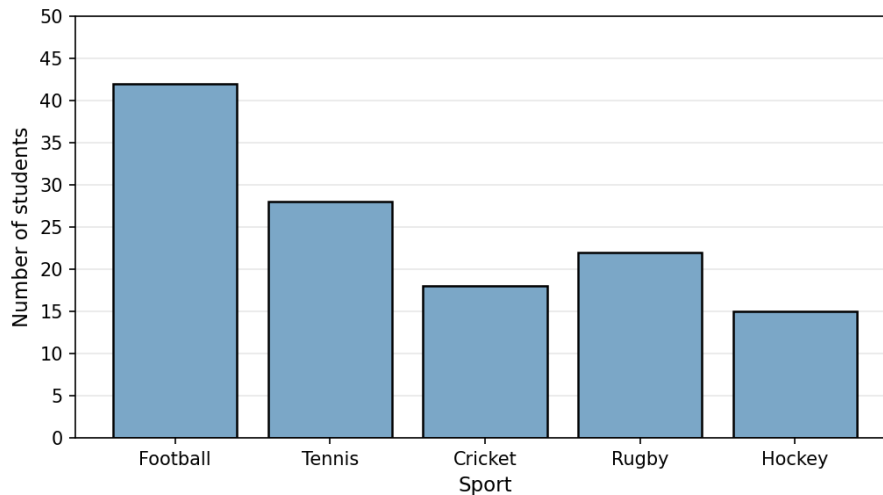
3. Work out $(6.4 \times 10^5) - (3.8 \times 10^4)$. Give your answer in standard form.

(3 marks)

4. Show that the recurring decimal 0.272727... (with the digits 27 repeating) can be written as the fraction $\frac{3}{11}$.

(3 marks)

5. The bar chart shows information about the favourite sports of 125 students.



(a) Write down the modal sport.

(1 marks)

(b) Calculate the percentage of students whose favourite sport is rugby. Give your answer to 1 decimal place.

(2 marks)

(c) One of the 125 students is chosen at random. Work out the probability that they chose cricket or hockey as their favourite.

(2 marks)

6. Solve $5(2x - 3) - 4(x + 1) = 6$.

(3 marks)

7. A box contains 7 blue pens and 3 red pens. Two pens are taken from the box at random, without replacement.

(a) Work out the probability that both pens are blue.

(3 marks)

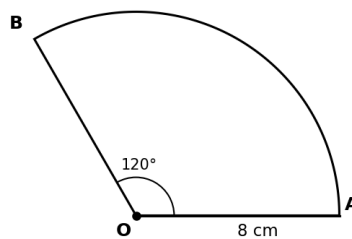
(b) Work out the probability that at least one pen is red.

(3 marks)

8. Line A has equation $2y + 3x = 8$. Line B is parallel to line A and passes through the point (6, 1). Find the equation of line B. Give your answer in the form $ax + by + c = 0$ where a, b, c are integers.

(4 marks)

9. The diagram shows a sector of a circle with centre O. The radius OA = 8 cm and angle AOB = 120° .



(a) Work out the length of arc AB. Give your answer in terms of π .

(2 marks)

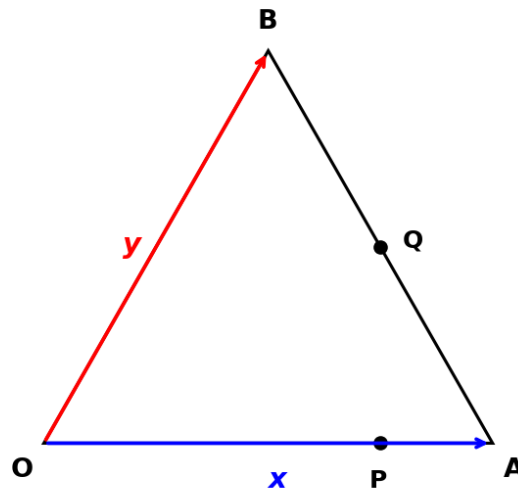
(b) Work out the area of the sector. Give your answer to 3 significant figures.

(3 marks)

10. Express $(7 + 5\sqrt{3}) / (1 + \sqrt{3})$ in the form $a + b\sqrt{3}$, where a and b are integers.

(4 marks)

11. OAB is a triangle. $\mathbf{OA} = \mathbf{x}$ and $\mathbf{OB} = \mathbf{y}$. P is the point on OA such that $OP : PA = 3 : 1$. Q is the midpoint of AB.



(a) Express $\mathbf{PQ}$ in terms of $\mathbf{x}$ and $\mathbf{y}$.

(4 marks)

(b) Hence prove that $\mathbf{PQ}$ is parallel to a vector of the form $(1/4)(p\mathbf{x} + q\mathbf{y})$ where p and q are integers to be found.

(2 marks)

12. Solve the simultaneous equations:

$$4x + y = 14$$

$$2x - 3y = -14$$

(3 marks)

13. Factorise fully:

(a) $x^2 - 49$

(1 marks)

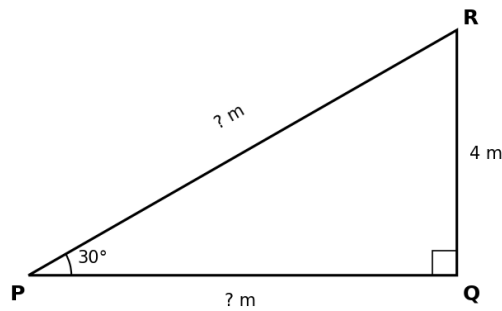
(b) $6x^2 + 11x - 10$

(2 marks)

(c) $2x^3 - 8x$

(2 marks)

14. A ramp PQR is in the shape of a right-angled triangle with the right angle at Q. PR rises at an angle of 30° from PQ. The height QR = 4 m.



(a) Work out the length of PR (the slope of the ramp). Give your answer to 3 significant figures.

(2 marks)

(b) Work out the length of PQ (the horizontal base). Give your answer to 3 significant figures.

(2 marks)

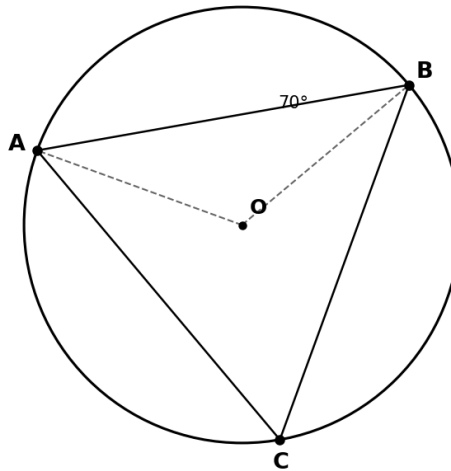
15. The product of two consecutive positive integers is 156. Form a quadratic equation and solve it to find the two integers.

(4 marks)

16. Write $\frac{3}{x-2} - \frac{2}{x+1}$ as a single fraction in its simplest form.

(3 marks)

17. A, B and C are points on the circumference of a circle, centre O. Angle AOB = 70° .



(a) Work out the size of angle OAB. Give a reason for your answer.

(2 marks)

(b) Work out the size of angle ACB. Give a reason for your answer.

(2 marks)

18. y is directly proportional to x and inversely proportional to the square of z . When $x = 5$ and $z = 2$, $y = 15$.

(a) Find a formula for y in terms of x and z .

(3 marks)

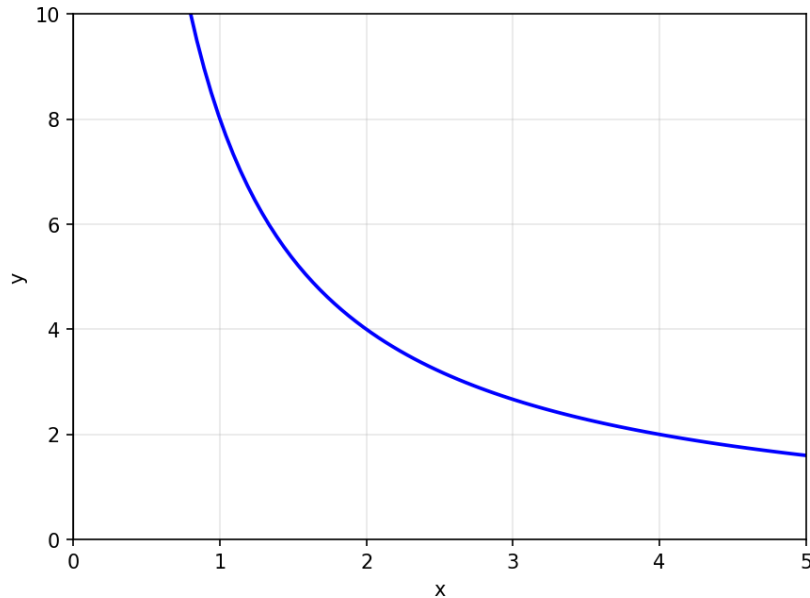
(b) Find the value of y when $x = 8$ and $z = 4$.

(2 marks)

19. Anna walks 3 km at a speed of x km/h, and then cycles 12 km at a speed of $(x + 6)$ km/h. Her total travel time is 1.5 hours. Form an equation in x and solve it to find Anna's walking speed.

(5 marks)

20. The graph of $y = 8/x$ is shown.



(a) Use the graph to estimate the value of y when $x = 3$.

(1 marks)

(b) By drawing a suitable straight line on the graph (or by algebra), find the solutions to the equation $8/x = x + 2$.

(3 marks)

21. A nanometre is 10^{-9} metres. A particular virus is approximately 120 nanometres in diameter.

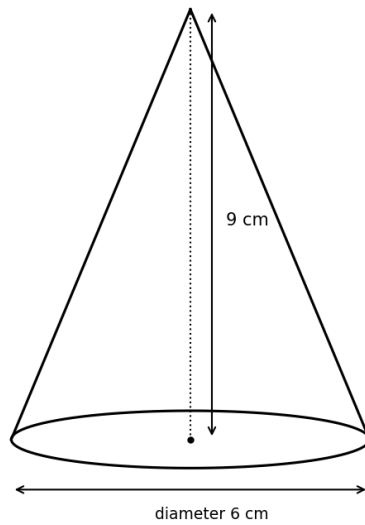
(a) Write the diameter of the virus in metres, in standard form.

(2 marks)

(b) Approximately how many such viruses, arranged in a straight line touching each other, would span 1 millimetre? Give your answer to 2 significant figures.

(3 marks)

22. The diagram shows a solid cone with diameter 6 cm and height 9 cm. A similar (smaller) cone is made from the same material with a diameter of 4 cm.



(a) Work out the volume of the original cone. Give your answer in terms of π .

(2 marks)

(b) Work out the volume of the smaller similar cone. Give your answer to 3 significant figures.

(3 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 2 Set E

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	£50.625 (= £50.63)	M1: 45×1.125 OR $45 + 0.125 \times 45$. A1: 50.625 (accept 50.63). [2]
2	Alice £120, Ben £180, Chen £240	M1: total parts = 9. M1: 1 part = $540/9 = 60$. A1: amounts in correct ratio (120, 180, 240). [3]
3	6.02×10^5	M1: rewrites $3.8 \times 10^4 = 0.38 \times 10^5$. M1: $6.4 - 0.38 = 6.02$. A1: 6.02×10^5 . [3]
4	3/11	M1: lets $x = 0.2727\dots$, so $100x = 27.2727\dots$. M1: $100x - x = 27 \Rightarrow 99x = 27$. A1: $x = 27/99 = 3/11$. [3]
5(a)	Football	B1: identifies football (42 students). [1]
5(b)	17.6%	M1: $22/125 \times 100$. A1: 17.6 (1 d.p.). [2]
5(c)	33/125	M1: cricket + hockey = $18 + 15 = 33$. A1: 33/125. [2]
6	$x = 25/6$	M1: expands: $10x - 15 - 4x - 4 = 6$. M1: $6x - 19 = 6 \Rightarrow 6x = 25$. A1: $x = 25/6$ (or 4.17). [3]
7(a)	$42/90 = 7/15$	M1: $(7/10) \times (6/9)$. A1: $42/90 = 7/15$. [3]
7(b)	$48/90 = 8/15$	M1: $P(\text{at least 1 red}) = 1 - P(\text{both blue})$. M1: $1 - 7/15$. A1: 8/15. [3]
8	$3x + 2y - 20 = 0$	M1: gradient of A = $-3/2$ (rearranges to $y = -(3/2)x + 4$). M1: substitutes (6, 1): $1 = -9 + c \Rightarrow c = 10$. M1: $y = -(3/2)x + 10$. A1: rearranges to $3x + 2y - 20 = 0$ (or equivalent integer form). [4]
9(a)	Arc length = $16\pi/3$ cm	M1: $(120/360) \times 2\pi(8)$. A1: $16\pi/3$ (accept 16.76 cm). [2]
9(b)	Area = 67.0 cm^2 (3 s.f.)	M1: $(120/360) \times \pi \times 8^2$. M1: $= 64\pi/3$. A1: 67.0 (3 s.f.). [3]
10	$4 + \sqrt{3}$	M1: multiplies top and bottom by conjugate $(1 - \sqrt{3})$. M1: numerator = $(7 + 5\sqrt{3})(1 - \sqrt{3}) = 7 - 7\sqrt{3} + 5\sqrt{3} - 15 = -8 - 2\sqrt{3}$. M1: denominator = $(1)^2 - (\sqrt{3})^2 = -2$. A1: $(-8 - 2\sqrt{3}) / -2 = 4 + \sqrt{3}$, so $a = 4$, $b = 1$. [4]
11(a)	$\mathbf{PQ} = -(1/4)\mathbf{x} + (1/2)\mathbf{y}$	M1: $\mathbf{OP} = (3/4)\mathbf{x}$. M1: $\mathbf{OQ} = (1/2)(\mathbf{x} + \mathbf{y}) = (1/2)\mathbf{x} + (1/2)\mathbf{y}$. M1: $\mathbf{PQ} = \mathbf{OQ} - \mathbf{OP}$. A1: $(1/2)\mathbf{x} + (1/2)\mathbf{y} - (3/4)\mathbf{x} = -(1/4)\mathbf{x} + (1/2)\mathbf{y}$. [4]
11(b)	$p = -1$, $q = 2$	M1: rewrites $-(1/4)\mathbf{x} + (1/2)\mathbf{y} = (1/4)(-\mathbf{x} + 2\mathbf{y})$. A1: $p = -1$, $q = 2$. [2]
12	$x = 2$, $y = 6$	M1: multiplies eq1 by 3: $12x + 3y = 42$. M1: adds to eq2: $14x = 28 \Rightarrow x = 2$. A1: substitutes to find $y = 14 - 8 = 6$. [3]
13(a)	$(x + 7)(x - 7)$	B1: correct factorisation. [1]
13(b)	$(3x - 2)(2x + 5)$	M1: identifies factors of $6 \times (-10) = -60$ summing to 11: 15 and -4. A1: $(3x - 2)(2x + 5)$. [2]
13(c)	$2x(x - 2)(x + 2)$	M1: takes out 2x: $2x(x^2 - 4)$. A1: factorises difference of squares to give $2x(x - 2)(x + 2)$. [2]
14(a)	PR = 8 m	M1: $\sin 30 = QR/PR \Rightarrow PR = 4 / \sin 30 = 4 / 0.5$. A1: 8. [2]
14(b)	PQ = 6.93 m (3 s.f.)	M1: $\tan 30 = QR/PQ \Rightarrow PQ = 4 / \tan 30$. A1: 6.93. [2]
15	12 and 13	M1: let integers be n and $n+1$; $n(n+1) = 156$. M1: forms $n^2 + n - 156 = 0$. M1: factorises $(n - 12)(n + 13) = 0$. A1: $n = 12$, so integers are 12 and 13. [4]
16	$(x + 7) / ((x - 2)(x + 1))$	M1: common denominator $(x - 2)(x + 1)$. M1: numerator = $3(x + 1) - 2(x - 2) = 3x + 3 - 2x + 4 = x + 7$. A1: $(x + 7) / ((x - 2)(x + 1))$. [3]
17(a)	Angle OAB = 55°	M1: OA = OB (radii), so triangle OAB is isosceles. Angles at A and B are equal. M1: angles sum to 180: $(180 - 70)/2 = 55$. A1: 55° with reason. [2]

Q	Answer	Mark scheme
17(b)	Angle ACB = 35°	B1: 35° . B1: reason "angle at centre is twice angle at circumference" ($70 / 2 = 35$). [2]
18(a)	$y = 12x/z^2$	M1: $y = kx/z^2$. M1: $15 = 5k/4 \Rightarrow k = 12$. A1: $y = 12x/z^2$. [3]
18(b)	$y = 6$	M1: $y = 12(8)/16$. A1: $y = 6$. [2]
19	$x = 6$ (walking speed = 6 km/h)	M1: time walking = $3/x$; time cycling = $12/(x + 6)$. M1: forms $3/x + 12/(x + 6) = 1.5$. M1: multiplies through to clear fractions: $6(x + 6) + 24x = 3x(x + 6)$. M1: simplifies to $x^2 - 4x - 12 = 0$. A1: factorises $(x - 6)(x + 2) = 0$; rejects $x = -2$; $x = 6$ km/h. [5]
20(a)	$y \approx 2.7$	B1: reads $y = 8/3 \approx 2.67$. [1]
20(b)	$x \approx 2$ or $x \approx -4$	M1: rearranges to $8 = x(x + 2) \Rightarrow x^2 + 2x - 8 = 0$. M1: factorises $(x - 2)(x + 4) = 0$. A1: $x = 2$ or $x = -4$. [3]
21(a)	1.2×10^{-7} m	M1: 120×10^{-9} . A1: 1.2×10^{-7} . [2]
21(b)	8300 viruses (2 s.f.)	M1: $1 \text{ mm} = 10^{-3} \text{ m}$. M1: $10^{-3} / (1.2 \times 10^{-7})$. A1: 8333... $\Rightarrow$ 8300 (2 s.f.). [3]
22(a)	$27\pi \text{ cm}^3$	M1: $V = (1/3)\pi r^2 h = (1/3)\pi(3)^2(9)$. A1: 27π . [2]
22(b)	25.1 cm^3 (3 s.f.)	M1: length scale factor = $4/6 = 2/3$. M1: volume scale factor = $(2/3)^3 = 8/27$. A1: $27\pi \times 8/27 = 8\pi \approx 25.1$. [3]

Worked Solutions — Predicted Paper 2 Set E

Higher Tier · Edexcel 1MA1 style

Q1: Service charge

Total = $45 \times 1.125 = \text{£}50.63$ (to nearest penny).

Or: service charge = $0.125 \times 45 = \text{£}5.625$, so total = $45 + 5.625 = \text{£}50.625$.

Q2: Ratio

Total parts = $2 + 3 + 4 = 9$. 1 part = $540 \div 9 = \text{£}60$.

Alice: $2 \times 60 = \text{£}120$. Ben: $3 \times 60 = \text{£}180$. Chen: $4 \times 60 = \text{£}240$.

Q3: Standard form subtraction

Rewrite to a common power: $3.8 \times 10^4 = 0.38 \times 10^5$.

$(6.4 - 0.38) \times 10^5 = \mathbf{6.02 \times 10^5}$.

Q4: Recurring decimal

Let $x = 0.272727\ldots$

Then $100x = 27.2727\ldots$

Subtract: $99x = 27 \Rightarrow x = 27/99 = \mathbf{3/11}$. ✓

Q5: Bar chart

(a) The tallest bar is Football, with 42 students. Modal sport: **Football**.

(b) Rugby: $22/125 \times 100 = \mathbf{17.6\%}$ (1 d.p.).

(c) Cricket + Hockey = $18 + 15 = 33$ students. Probability = $\mathbf{33/125}$.

Q6: Equation with brackets

Expand: $10x - 15 - 4x - 4 = 6$.

Simplify: $6x - 19 = 6 \Rightarrow 6x = 25 \Rightarrow \mathbf{x = 25/6}$ (or 4.17 to 2 d.p.).

Q7: Probability without replacement

(a) $P(\text{both blue}) = (7/10) \times (6/9) = 42/90 = \mathbf{7/15}$.

(b) $P(\text{at least 1 red}) = 1 - P(\text{both blue}) = 1 - 7/15 = \mathbf{8/15}$.

Q8: Parallel line

Rearrange Line A: $2y = -3x + 8 \Rightarrow y = -(3/2)x + 4$. Gradient = $-3/2$.

Line B has the same gradient. Substitute (6, 1): $1 = -(3/2)(6) + c \Rightarrow c = 10$.

So $y = -(3/2)x + 10$. Multiply by 2: $2y = -3x + 20$.

Rearrange: $\mathbf{3x + 2y - 20 = 0}$.

Q9: Sector

(a) Arc length = $(120/360) \times 2\pi r = (1/3) \times 16\pi = \mathbf{16\pi/3 \text{ cm}}$ ($\approx 16.8 \text{ cm}$).

(b) Sector area = $(120/360) \times \pi r^2 = (1/3) \times 64\pi = \mathbf{64\pi/3 \text{ cm}^2}$.

$$= 67.0 \text{ cm}^2 \text{ (3 s.f.)}.$$

Q10: Rationalising

Multiply top and bottom by the conjugate $(1 - \sqrt{3})$:

$$\text{Numerator: } (7 + 5\sqrt{3})(1 - \sqrt{3}) = 7 - 7\sqrt{3} + 5\sqrt{3} - 5(\sqrt{3})^2$$

$$= 7 - 2\sqrt{3} - 15 = -8 - 2\sqrt{3}.$$

$$\text{Denominator: } (1 + \sqrt{3})(1 - \sqrt{3}) = 1 - 3 = -2.$$

$$\text{Result: } (-8 - 2\sqrt{3}) / (-2) = 4 + \sqrt{3}.$$

So $a = 4$, $b = 1$.

Q11: Vectors

$$\text{(a) } \mathbf{OP} = (3/4)\mathbf{OA} = (3/4)\mathbf{x}.$$

$$\mathbf{OQ} = (1/2)(\mathbf{OA} + \mathbf{OB}) = (1/2)\mathbf{x} + (1/2)\mathbf{y}.$$

$$\mathbf{PQ} = \mathbf{OQ} - \mathbf{OP} = (1/2)\mathbf{x} + (1/2)\mathbf{y} - (3/4)\mathbf{x} = -(1/4)\mathbf{x} + (1/2)\mathbf{y}.$$

$$\text{(b) Rewrite: } -(1/4)\mathbf{x} + (1/2)\mathbf{y} = (1/4)(-\mathbf{x} + 2\mathbf{y}).$$

So $p = -1$, $q = 2$.

Q12: Simultaneous equations

$$4x + y = 14 \dots (1)$$

$$2x - 3y = -14 \dots (2)$$

$$\text{Multiply (1) by 3: } 12x + 3y = 42 \dots (3)$$

$$\text{Add (2) and (3): } 14x = 28 \Rightarrow x = 2.$$

$$\text{Substitute into (1): } 4(2) + y = 14 \Rightarrow y = 6.$$

$$\mathbf{x = 2, y = 6.}$$

Q13: Factorising

$$\text{(a) Difference of squares: } x^2 - 49 = (\mathbf{x + 7})(\mathbf{x - 7}).$$

$$\text{(b) } 6x^2 + 11x - 10. \text{ Find factors of } 6 \times -10 = -60 \text{ summing to 11: 15 and -4.}$$

$$6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5) = (\mathbf{3x - 2})(\mathbf{2x + 5}).$$

$$\text{(c) } 2x^3 - 8x = 2x(x^2 - 4) = \mathbf{2x(x - 2)(x + 2)}.$$

Q14: Ramp trig

$$\text{(a) } \sin 30^\circ = \text{QR/PR} = 4/\text{PR} \Rightarrow \text{PR} = 4 / \sin 30^\circ = 4 / 0.5 = \mathbf{8 \text{ m}}.$$

$$\text{(b) } \tan 30^\circ = \text{QR/PQ} = 4/\text{PQ} \Rightarrow \text{PQ} = 4 / \tan 30^\circ = 4 / 0.5774 = \mathbf{6.93 \text{ m (3 s.f.)}}.$$

Q15: Quadratic word problem

Let the two consecutive integers be n and $n + 1$.

$$n(n + 1) = 156 \Rightarrow n^2 + n - 156 = 0.$$

$$\text{Factorise: } (n - 12)(n + 13) = 0 \Rightarrow n = 12 \text{ or } n = -13.$$

Since we need positive integers, $n = 12$. So the integers are **12 and 13**.

Check: $12 \times 13 = 156$. ✓

Q16: Algebraic fractions

Common denominator = $(x - 2)(x + 1)$.

$$(3) / (x - 2) - (2) / (x + 1) = [3(x + 1) - 2(x - 2)] / ((x - 2)(x + 1)).$$

Numerator: $3x + 3 - 2x + 4 = x + 7$.

Answer: $(x + 7) / ((x - 2)(x + 1))$.

Q17: Circle (centre angle, isosceles)

(a) Triangle OAB is isosceles since $OA = OB$ (both radii).

Angles at A and B are equal. Sum of triangle angles = 180° .

Angle OAB = $(180 - 70) / 2 = 55^\circ$ (base angles of isosceles triangle).

(b) By the 'angle at the centre is twice angle at the circumference' theorem,

angle ACB = $(1/2) \times$ angle AOB = $(1/2) \times 70 = 35^\circ$.

Q18: Direct and inverse proportion

(a) $y = kx/z^2$. Substitute $x = 5$, $z = 2$, $y = 15$: $15 = 5k/4 \Rightarrow k = 12$.

$$y = 12x/z^2.$$

(b) $y = 12(8)/16 = 96/16 = 6$.

Q19: Speed-distance equation

Time walking = $3/x$. Time cycling = $12/(x + 6)$. Total time = 1.5 hours.

$$3/x + 12/(x + 6) = 1.5.$$

Multiply through by $2x(x + 6)$ (= twice the common denominator):

$$6(x + 6) + 24x = 3x(x + 6).$$

$$6x + 36 + 24x = 3x^2 + 18x.$$

$$30x + 36 = 3x^2 + 18x \Rightarrow 3x^2 - 12x - 36 = 0.$$

$$\text{Divide by 3: } x^2 - 4x - 12 = 0.$$

$$\text{Factorise: } (x - 6)(x + 2) = 0 \Rightarrow x = 6 \text{ or } x = -2.$$

Speed must be positive: $x = 6 \text{ km/h}$.

(So Anna walks at 6 km/h and cycles at 12 km/h.)

Q20: Reciprocal graph

(a) When $x = 3$, $y = 8/3 \approx 2.67$ (or 2.7 to 1 d.p.).

(b) $8/x = x + 2$. Multiply both sides by x : $8 = x(x + 2) = x^2 + 2x$.

Rearrange: $x^2 + 2x - 8 = 0$. Factorise: $(x - 2)(x + 4) = 0$.

So $x = 2$ or $x = -4$.

Q21: Standard form / unit conversion

(a) Diameter = 120 nm = $120 \times 10^{-9} \text{ m} = 1.2 \times 10^{-7} \text{ m}$.

(b) 1 mm = 10^{-3} m . Number of viruses = $(10^{-3}) / (1.2 \times 10^{-7})$.

$$= (1/1.2) \times 10^{-3 - (-7)} = 0.8333 \times 10^4 = 8333.$$

To 2 s.f.: **8300 viruses**.

Q22: Cone volume + similarity

(a) Radius = 3 cm. $V = (1/3)\pi r^2 h = (1/3)\pi \times 9 \times 9 = 27\pi \text{ cm}^3$.

(b) Length scale factor (small : large) = 4 : 6 = 2 : 3.

Volume scale factor = $(2/3)^3 = 8/27$.

Volume of smaller cone = $27\pi \times 8/27 = 8\pi = 25.1 \text{ cm}^3$ (3 s.f.).